Mapping the Cultural Fingerprints of LLM's

Background

AI, specifically LLM's, are a very big topic all over the world right now. People generally praise how flexible these models are and how they can do almost anything. There are dozens if not hundreds of different AI companies each with their own model(s) that are trying to sell to consumers that their AI model should be used over their competitors. Now besides public sentiment, how does one actually determine which AI model is better than the others? Thats where benchmarks come into play. Benchmarks are to LLM's as standardized tests are to students. Essentially people have created different benchmarks for the LLMs to determine how good a model is at a specific task (think specific AP tests to evaluate students capabilities on a subject). Now the typical use case of these benchmarks (by the public and by companies deciding which company to buy enterprise accounts for) typically analyze which models or AI company tends to be top performers at the benchmarks for their use cases. Noow typically people only look at how models perform on a specific benchmark or small set of benchmarks to determine which model is the best. But what if the constant creation of new models and benchmarks have underlying implications and features people have yet to notice?

What is the Project

My project studies AI benchmark performance in aggregate, rather than focusing only on top-performing models. I intervene by asking: what underlying factors shape what we call "performance"? While many people assume LLM benchmarks are assumed to be neutral leaderboards, I question this assumption by studying patterns that emerge from mapping LLM models onto a coordinate space, based on their benchmark performance, where models that perform similarly across benchmarks cluster together. I hope to find whether regional origin predicts performance patterns and whether domain strengths form distinct cognitive styles. If so, this changes the landscape: benchmarks become cultural artifacts rather than neutral yardsticks; AI evaluation becomes a site of cultural negotiation; and "intelligence" must be understood as plural, situated, and political.

Separation From Other Humanities Works

In HTEC301, we read this quantamagazine article on how AI models converge on how they encode reality. The TLDR of the article is that the way AI models seem to understand words are phrases seem to be converging to some similar shared representation. Now while this is a similar study and inspiration to my work, I see my work and theirs as quite different. Imagine a study that analyzes how students of different grade levels define different words and find synonyms, this study would likely find that as kids are older and their understanding of words increase the students would likely have similar synonyms/definitions of words and their vocabulary would be broader. This study is very similar to what the quantamagazine cited study is about. My work is more akin to having these same students write essays on various topics and analyze what emergent properties come into play on which topics do students perform similarly in, are we able to form clusters of students based on essay strengths, etc. Now the way I approach this is through representing AI models as a vector formed through their benchmark scores and performing qualitative analysis of how the models cluster. To do this qualitative analysis I employ various dimensionality reduction techniques like PCA and UMAP which we can think of for now as a way to view high dimensional vectors in a more digestable 2 or 3 dimensional way (more explanation in class).

Significance

Benchmarks are quietly making cultural policy since they determine which models get funded, deployed, and regulated. If a benchmark encodes one tradition's definition of "correct reasoning," then evaluation is

a mechanism of cultural control. Additionally my project challenges the idea of the "AI Race" metaphor. If models from different origins develop distinct cognitive profiles (math+logic vs math+reasoning), then we are not watching a single ladder being climbed, rahter divergent intellectual traditions being compressed into one ranking. If every lab optimizes toward the same benchmark(s), diversity of problem-solving style erodes, whats worse is that erosion is labelled progress.

Current State

https://aidanbugs.github.io/Cluster-Culture-AI-Benchmark-Explorer/

Fall	Spring
(+) Benchmarking Data	(~) In depth data analysis
(+) Cool Data Plots	(_) Insights towards significance and impact
(+) Website [its hosted! :)]	(_) Essay on findings
(_) User friendly experience	

Feedback

Does the project make sense? What can I do to better frame it? Is there anything I need to clarify / keep in mind when I start my essay?

Does the significance feel impactful to you? Does it feel like I'm missing something? What more can I try to extract from this data/project?